

REQUIRED DELIVERABLE

Scoring Methodology

Explainable multi-dimensional scoring model for COMPASSIQ.

Version: Final · Status: Documentation-ready

Project: AINS Hackathon 2026 — AI for Entrepreneurship

Team: Starlight

Members: Senda Selmi · Yoser Ladhibi

Contacts: sendaselmi111@gmail.com · yoserladhibi@gmail.com

Prepared for GitHub repository submission

1. Objective

The scoring methodology provides an explainable, multi-dimensional assessment of a startup's entrepreneurial readiness. Rather than assigning a single opaque score, COMPASSIQ evaluates five complementary dimensions reflecting the main pillars of startup development.

Each dimension is computed from measurable sub-criteria, controlled by gating rules, explained in plain language, and linked to concrete improvement recommendations.

2. Design Principles

Principle	Description
Explainability	Every score traces back to specific questionnaire responses and project evidence.
Transparency	Weights and aggregation logic are predefined and publicly documented.
Actionability	Every low score triggers a concrete improvement recommendation.
Independence	Each dimension evaluates a distinct entrepreneurial capability.
Fairness	Projects are compared against objective criteria, not subjective opinions.
Non-masking	Composite scores are not simple averages: a weak fundamental dimension cannot be hidden by strong scores elsewhere.

3. Composite Score Architecture

Dimension	Weight in Overall Score
Market Score	25%
Commercial Offer Score	20%
Innovation Score	20%
Scalability Score	20%
Green Score	15%

4. Market Score (0-100)

Purpose: Measures evidence of real market demand, customer validation, and revenue viability.

Criterion	Weight	What is measured
Customer Validation Evidence	35%	Number and quality of interviews, surveys, LOIs or pilot users demonstrating real demand.
Market Share Analysis	35%	Addressable market size (TAM/SAM/SOM) combined with competitive landscape mapping.
Revenue Model Clarity & Viability	30%	Clarity of the monetization path and realism of projected revenue.

Gating Rule: If customer interviews < 5 → Market Score capped at 50.

Score Range	Meaning
75-100	Strong validation evidence, clear and viable revenue path.
50-74	Partial validation, revenue model in progress.
0-49	Limited market evidence — fundamental gaps remain.

5. Commercial Offer Score (0-100)

Purpose: Measures the commercial coherence, maturity, and readiness of the product or service.

Criterion	Weight	What is measured
Value Proposition Clarity & Differentiation	30%	How clearly the offer is articulated and how it stands apart from competitors.
Product/Service Maturity & Readiness	25%	Development stage: concept → prototype → MVP → market-ready.
Pricing Strategy Coherence	25%	Pricing logic, margin awareness, and alignment with customer willingness to pay.
Offer-Customer Alignment	20%	Degree to which the offer directly addresses identified customer needs.

Gating Rule: If revenue = 0 AND paying customers = 0 → Commercial Offer Score capped at 50.

Score Range	Meaning
75-100	Product is commercially coherent and market-ready.
50-74	Offer is defined but commercial readiness is incomplete.
0-49	Offer requires fundamental redesign or further validation.

6. Innovation Score (0-100)

Purpose: Evaluates uniqueness, technological value, and defensibility of the solution in the local and regional context.

Criterion	Weight	What is measured
Local Novelty & Differentiation	30%	Degree of innovation specifically within the Tunisian/MENA context.
Technology Intensity	25%	Level of technology integration and sophistication applied to the solution.
Barrier to Entry	25%	Difficulty for a competitor to replicate or substitute the solution.
Departure from Existing Offerings	20%	How significantly the solution moves beyond current market alternatives.

Gating Rule: If prototype = No → Innovation Score capped at 60.

Score Range	Meaning
75-100	Highly differentiated, defensible, and technology-driven.
50-74	Moderate innovation with meaningful differentiation.
0-49	Limited novelty — solution closely mirrors existing offerings.

7. Scalability Score (0-100)

Purpose: Measures the startup's capacity to grow efficiently without proportional increase in cost or manual effort.

Criterion	Weight	What is measured
Replicability without Linear Cost Increase	30%	Can the solution serve 10x more users without 10x the cost?
Manual Accompaniment Dependency (inverted)	25%	Lower human-intervention dependency means higher score; heavy manual processes are penalized.
Deployment Cost Structure	20%	Balance of fixed vs. variable costs and infrastructure portability across geographies.
Geographic & Market Expansion Potential	25%	Feasibility of expansion beyond the initial geography.

Gating Rule: If growth plan = Missing → Scalability Score capped at 60.

Score Range	Meaning
75-100	Business model is structurally scalable.
50-74	Scalability is achievable but requires operational restructuring.
0-49	Growth is highly dependent on manual effort or narrow geography.

8. Green Score (0-100)

Purpose: Evaluates the startup's environmental footprint and responsibility across four pillars, aligned with Tunisian priorities and sustainability frameworks.

Pillar	Weight	What is measured
Climate & Air	30%	Carbon footprint, GHG emissions, climate impact, and mitigation measures.
Water	25%	Water consumption efficiency, impact on water quality, reuse and recycling practices.
Soil & Biodiversity	20%	Land use impact, ecosystem preservation, soil health, and biodiversity footprint.
Resources & Waste	25%	Resource efficiency, waste reduction, circular economy practices, and material lifecycle.

Gating Rule: If no environmental impact assessment is conducted AND sector is high-impact → Green Score capped at 40.

Score Range	Meaning
75-100	Strong environmental responsibility across all four pillars.
50-74	Partial sustainability practices — some pillars unaddressed.
0-49	Significant environmental concerns — sustainability not yet integrated.

SDG Alignment for Green Score

Pillar	Primary SDG
Climate & Air	SDG 13 — Climate Action
Water	SDG 6 — Clean Water & Sanitation
Soil & Biodiversity	SDG 15 — Life on Land
Resources & Waste	SDG 12 — Responsible Consumption & Production

9. Gating Rules — Complete Summary

Rule	Condition	Cap Applied	Affected Score
G1	customer interviews < 5	≤ 50	Market
G2	revenue = 0 AND paying customers = 0	≤ 50	Commercial Offer
G3	prototype = No	≤ 60	Innovation
G4	growth plan = Missing	≤ 60	Scalability
G5	no environmental assessment AND high-impact sector	≤ 40	Green

10. Global Score Interpretation

Score Range	Label	Dashboard Indicator
90–100	Excellent	Green
75–89	Strong	Green
60–74	Good	Yellow
40–59	Needs Improvement	Orange
0–39	Critical Priority	Red

11. Explainability Layer

Every score output must answer three questions:

- Why did the project receive this score?
- What is the primary gap?
- What concrete action would improve it?

Example Output

Market Score: 72/100

- ✓ **Strong customer interviews (15 conducted)**
- ✓ **Large addressable market (TAM estimated at \$50M)**
- ✗ **Revenue model not yet validated with paying customers**

Primary gap: Revenue Model Clarity & Viability (45/100)

Recommendation: Run a pricing pilot with 3–5 early adopters before the next fundraising conversation.

12. Diagnosis Confidence Score

Signal	Weight
Questionnaire completeness (% of questions answered)	50%
Presence of validation evidence (documents, interviews, revenue data)	30%
Response consistency (no contradictory signals detected)	20%

Dashboard Output Example

Dimension	Score
Market Score	72/100
Commercial Offer Score	58/100
Innovation Score	88/100

Dimension	Score
Scalability Score	65/100
Green Score	71/100
Diagnosis Confidence	84%

13. Example Calculation — Market Score

Criterion	Raw Score (0-100)	Weight
Customer Validation Evidence	80	35%
Market Share Analysis	90	35%
Revenue Model Clarity & Viability	60	30%

Weighted calculation:
Market Score = (80 × 0.35) + (90 × 0.35) + (60 × 0.30)
= 28 + 31.5 + 18 = 77.5 → rounded to 78/100

Gating check: Customer interviews conducted = 8 ≥ 5 → No cap applied.
Final Market Score: 78/100

14. Integration with the AI Assistant

The scoring engine passes a structured JSON payload to the LLM, which transforms it into natural-language guidance. This separation ensures the system explains — it does not invent.

Payload Example

```
{ "maturity_stage": "Structuration", "market_score": 78, "commercial_score": 58, "innovation_score": 88, "scalability_score": 65, "green_score": 71, "diagnosis_confidence": 84, "main_blocker": "Commercial Offer — Revenue model not validated", "gating_rules_triggered": ["G2"] }
```

Document version: 1.0 — AINS Hackathon 2026 — CompassIQ